

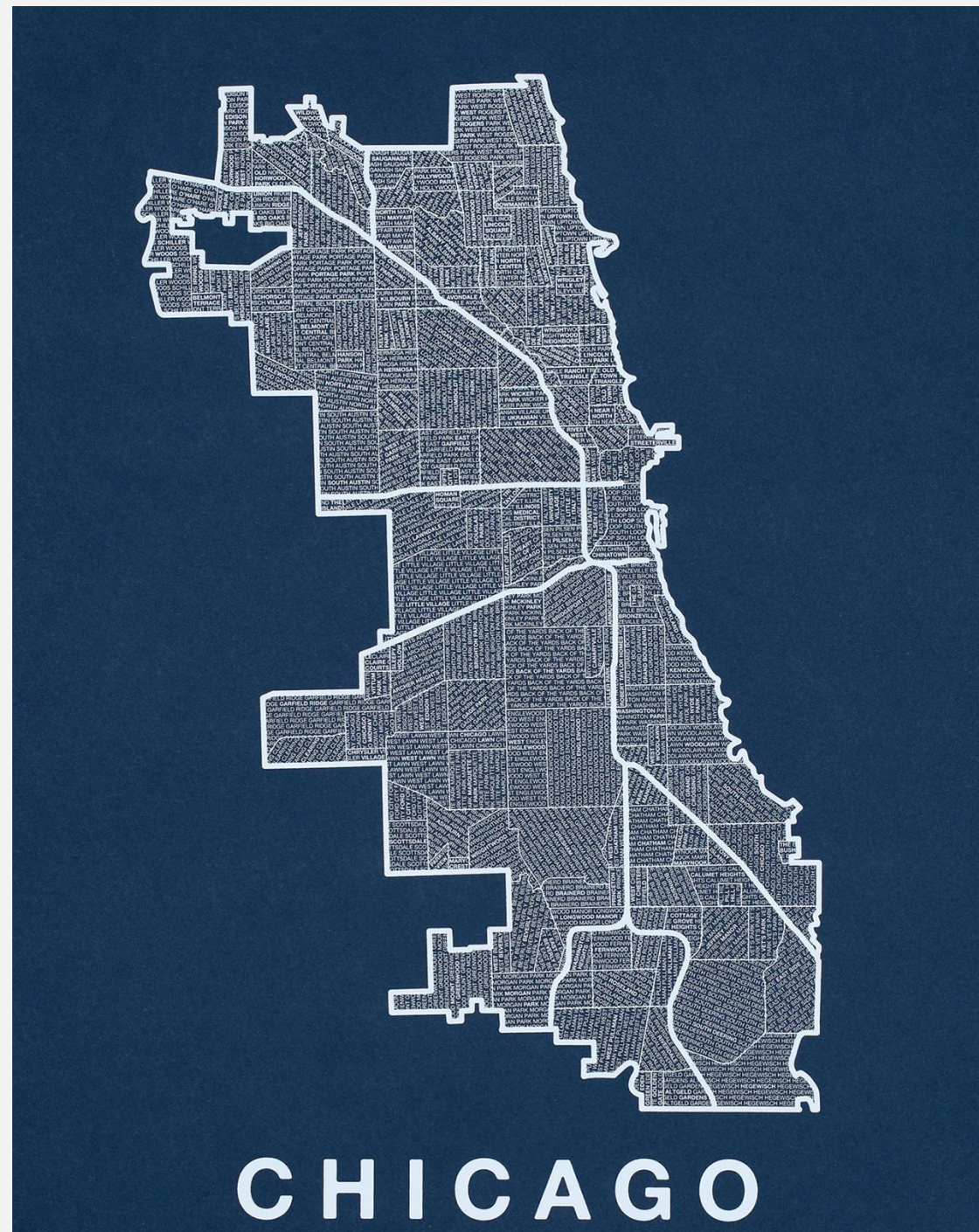
Modeling Solar Site Suitability in Chicago using GIS

ENV 756 Data Automation Project Report

Christian Dadzie

Section 1

Background



This project leverages GIS modeling to automate the identification of optimal community areas for renewable energy installations in Chicago. By integrating energy demand, socioeconomic data, and land use classifications, the model pinpoints the best locations for solar implementation across two land use types: Planned Development and Open Space/Residential zones.

This analysis is critical for addressing disparities in energy access, where high energy burdens and limited renewable installations disproportionately affect underserved communities.

Data Sources



The boundary and energy data were sourced from the Chicago Data Portal whereas the demographic data were sourced from the CMAP Data Portal

Boundary Data



- › Chicago Community Areas (Shapefile)
- › Land Use Inventory (Shapefile)

Demographic Data



- › Population (csv)
- › Economic Indicators (csv)
- › Race (csv)

Energy Data



- › Electricity Consumption (csv)
- › Natural Gas Consumption (csv)
- › Chicago Energy Benchmark Data(csv)

Section 2

The Supermodel

Geoprocessing

←

Supermodel

+

Parameters

Environments

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Chicago Community Areas

Chicago Community Areas

▼

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Diversity Analysis

DiversityAnalysis

▼

📁

Hardship Analysis

HardshipAnalysis

▼

📁

Suitable Zones: Open & Residential Areas

Suitable Zones:SuitableZones_4

▼

📁

Suitable Zones: Planned Development

Suitable Zones:SuitableZones_6

▼

📁

Sorted Total Score: Open/Res

Join_OpRes_Sort

📁

Sorted Total Score: PlannedDev

Join_PlannedDev_Sort

📁

▶ Run

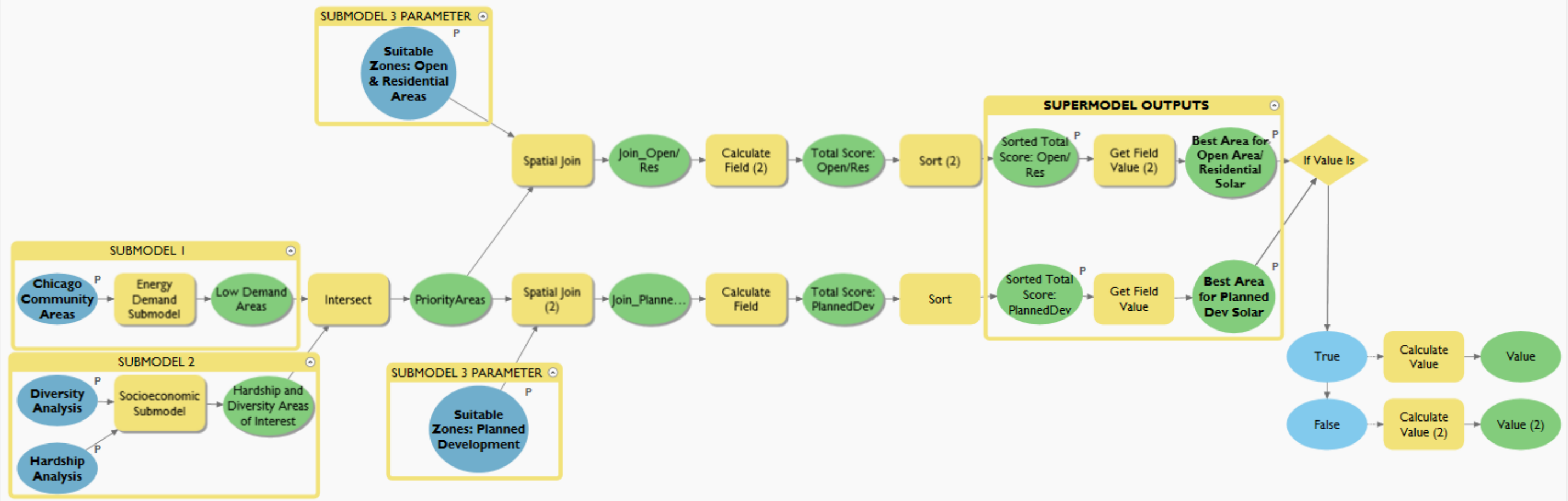
▼

Overview

The supermodel integrates three main submodels to identify optimal community areas for equitable renewable energy development:

- **Energy Demand Submodel:** Identifies energy consumption hotspots by normalizing electricity and gas usage across Chicago's community areas.
- **Socioeconomic Submodel:** Analyzes population-weighted hardship and diversity indices to prioritize areas with significant socioeconomic challenges.
- **Land Use and Efficiency Submodel:** Evaluates land use suitability and overlaps low-efficiency zones with strategic land classifications (Planned Development and Open Space/Residential).

Overview



Section 3

Energy Demand – Submodel 1

Geoprocessing

←

Energy Demand Submodel

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Parameters

Environments

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Chicago Community Areas

Chicago Community Areas

▼

Electricity Weight

0.6

Gas Weight

0.4

Low Demand Areas

LowDemandAreas

▼

Energy Demand Escalator

0.17

▶

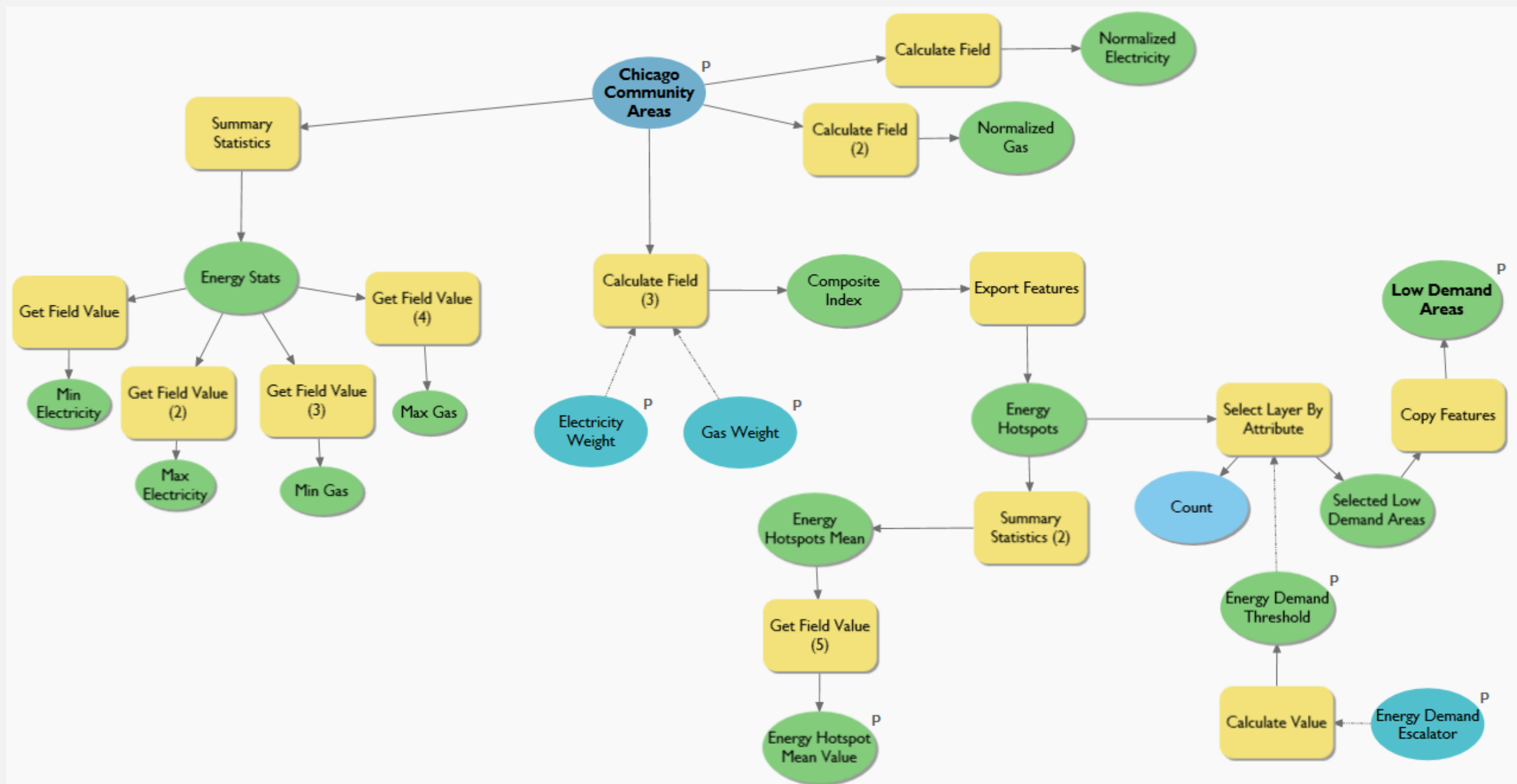
Run

▼

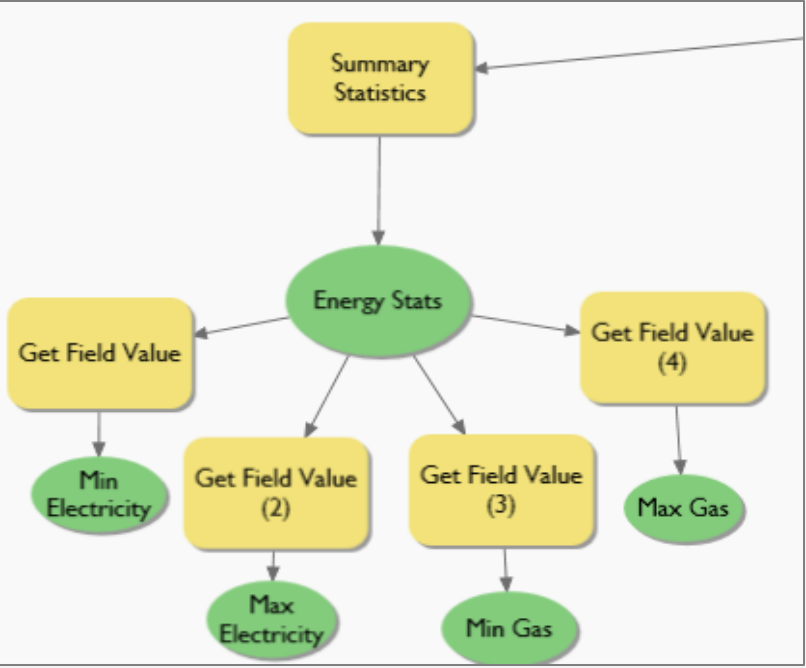
Overview

The Energy Demand Submodel identifies low-demand areas by **normalizing electricity** and **natural gas consumption** data on a **0–100 scale**, weighted at **60%** and **40%**, respectively, to generate a **Composite Demand Index**. Using a demand threshold, the model selects community areas with the lowest energy demand based on a **user-defined escalation factor**.

Overview



Key Features



MIN_EconomicIndicators_ExportTable1.KWH_TOTAL_SQFT	MAX_EconomicIndicators_ExportTable1.KWH_TOTAL_SQFT
9943.32	1154623.23

Min Electricity: Any Value

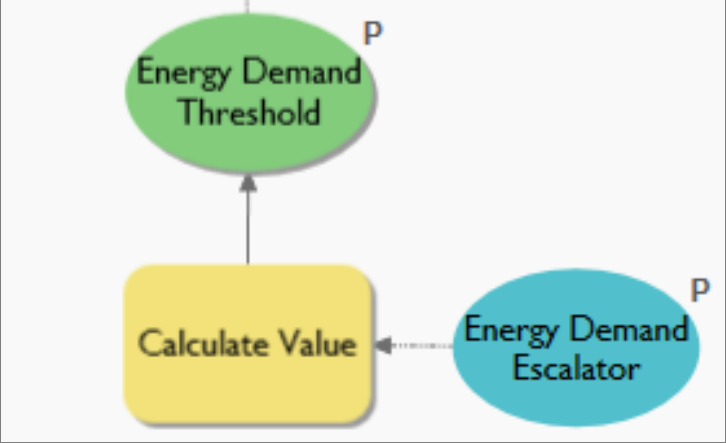
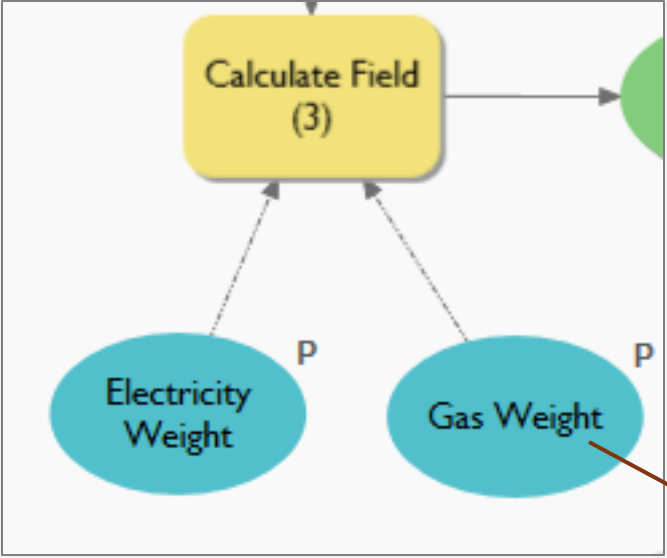
Min Electricity

9943.32

NormElec =

$$\frac{(!\text{EconomicIndicators_ExportTable1.KWH_TOTAL_SQFT!} - \% \text{Min Electricity\%})}{(\% \text{Max Electricity\%} - \% \text{Min Electricity\%})} * 100$$

Used **Get Field Value** to extract min and max values from **Summary Statistics** table to automate Normalization within the **Calculate Field** tool.



Electricity Weight	0.6
Gas Weight	0.4
Low Demand Areas	LowDemandAreas
Energy Demand Escalator	0.17

Gas Weight: Double

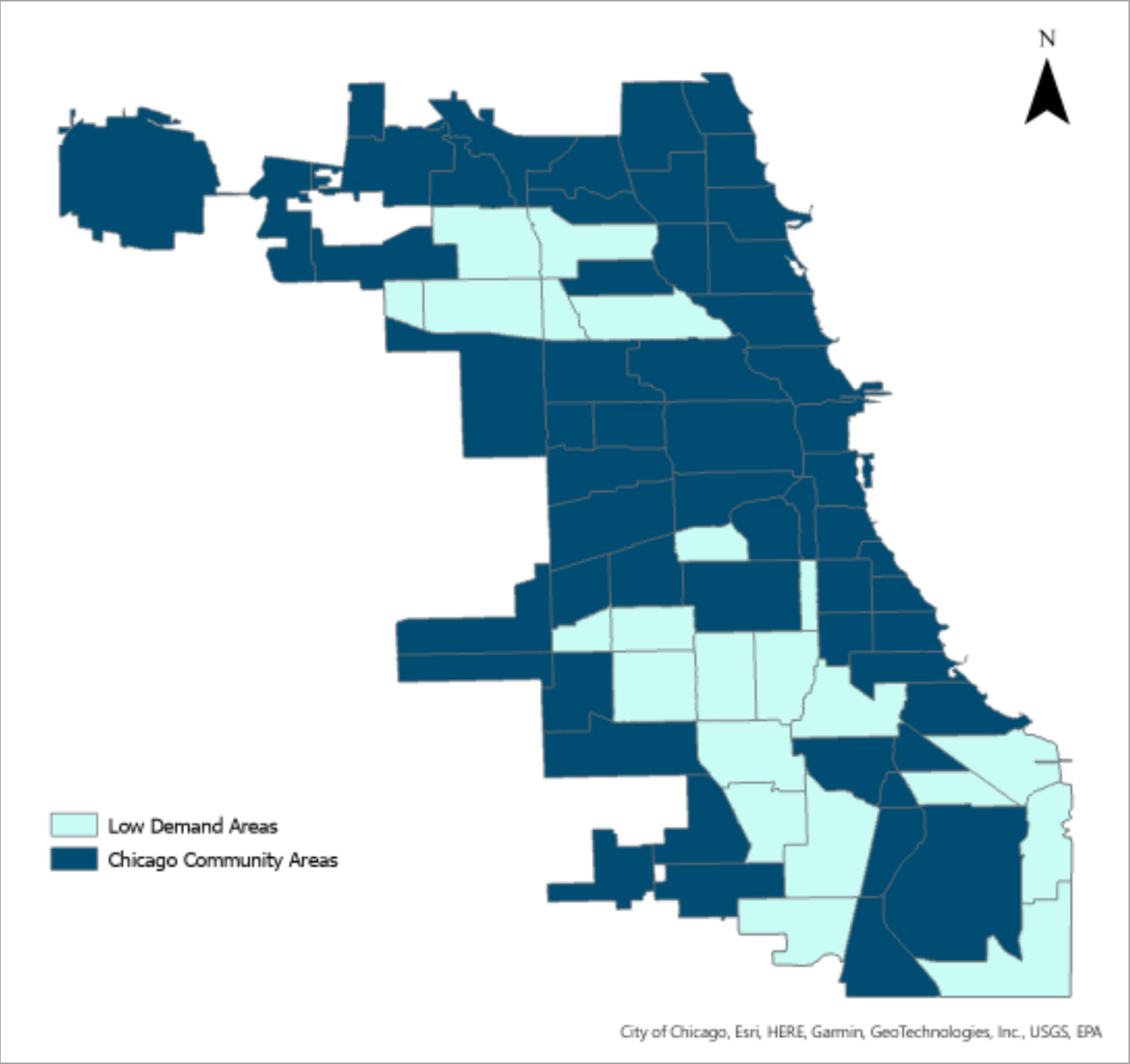
Gas Weight

0.4

Used **Create Variable** to produce a new variable and set them as **Parameters** to allow dynamic inputs for testing different threshold values efficiently.

Outputs

Normalized Electricity (kWh)	Normalized Gas (Therms)	Composite Index
0.746082	0.738777	0.74316
0.570693	0.661671	0.607084
0.593287	0.651546	0.616591
6.626786	5.617441	6.223048
0.674254	0.642773	0.661662
2.9623	3.186092	3.051817
0.950198	0.998493	0.969516
0.829009	1.06771	0.924489
29.382694	28.276812	28.940341
1.316675	1.370705	1.338287
0.796965	0.789388	0.793934
20.404607	18.798284	19.762078
100	100	100
1.258321	1.455513	1.337198
0.6611	0.653611	0.658105
0.479885	0.668028	0.555143



Section 3

Socioeconomic – Submodel 2

Geoprocessing

← Socioeconomic Submodel →

Parameters

Environments

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Diversity Analysis

DiversityAnalysis

Hardship and Diversity Areas of Interest

HardshipandDiversity

Hardship Index Escalator

1.35

Hardship Analysis

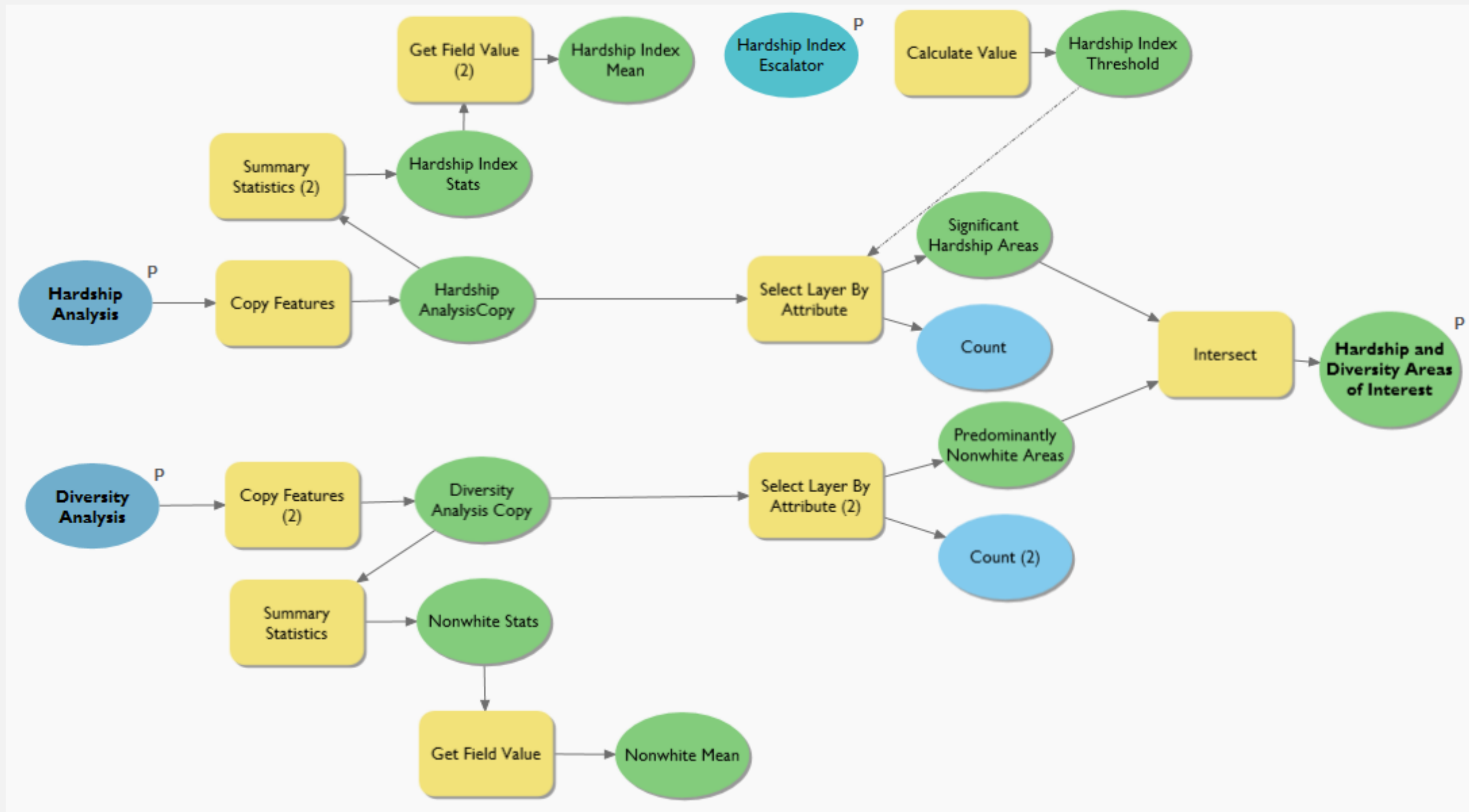
HardshipAnalysis

▶ Run ▼

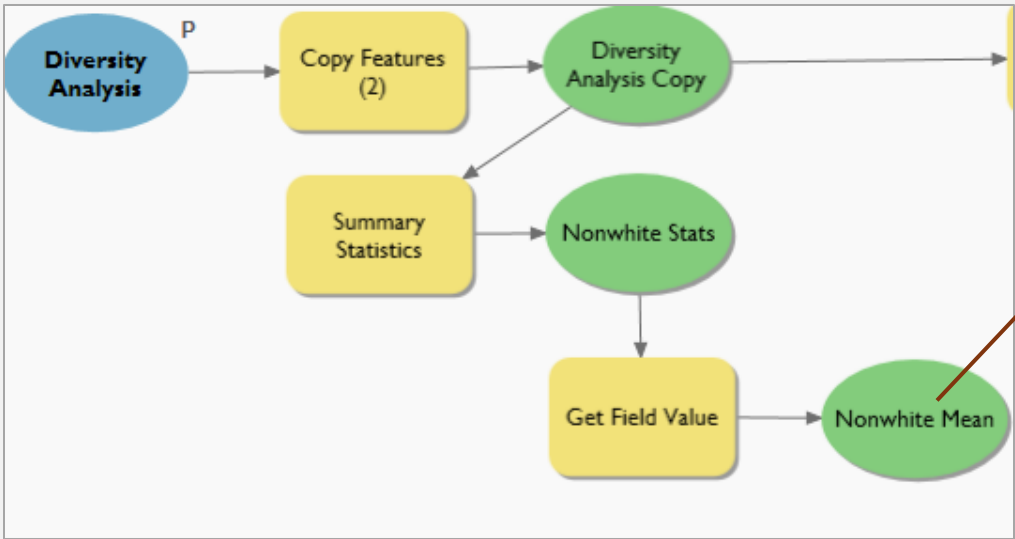
Overview

The Socioeconomic Submodel focuses on identifying areas with **high social and economic vulnerability**. It starts by analyzing hardship and diversity data separately. Using **summary statistics**, it calculates thresholds to define areas with significant hardship and predominantly non-white populations. These thresholds are then used to select key community areas. Finally, the model **intersects** these selected areas to pinpoint regions where social and economic challenges overlap.

Overview



Key Features

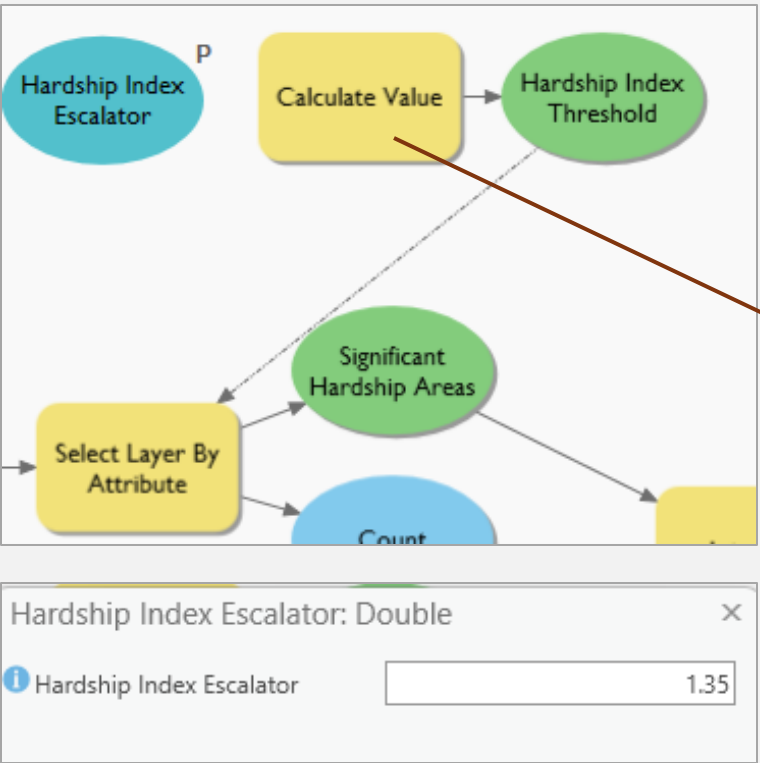


Nonwhite Mean: Any Value	
Nonwhite Mean	
73	

OBJECTID *	FREQUENCY	MEAN_Pop_HardIx
1	77	1563272.402597

Hardship Index Mean: Any Value	
Hardship Index Mean	
1563272	

Used **Get Field Value** to extract mean values from Hardship and Non-white population **Summary Statistics** for generating selection thresholds.



Pop_HardIx > %Hardship Index Threshold%

PctNonWhit > %Nonwhite Mean%

Expression

%Hardship Index Mean% * %Hardship Index

Select Layer By Attribute

Parameters Environments Properties

Input Rows

Hardship AnalysisCopy

Selection Type

New selection

Expression

Load Save Remove

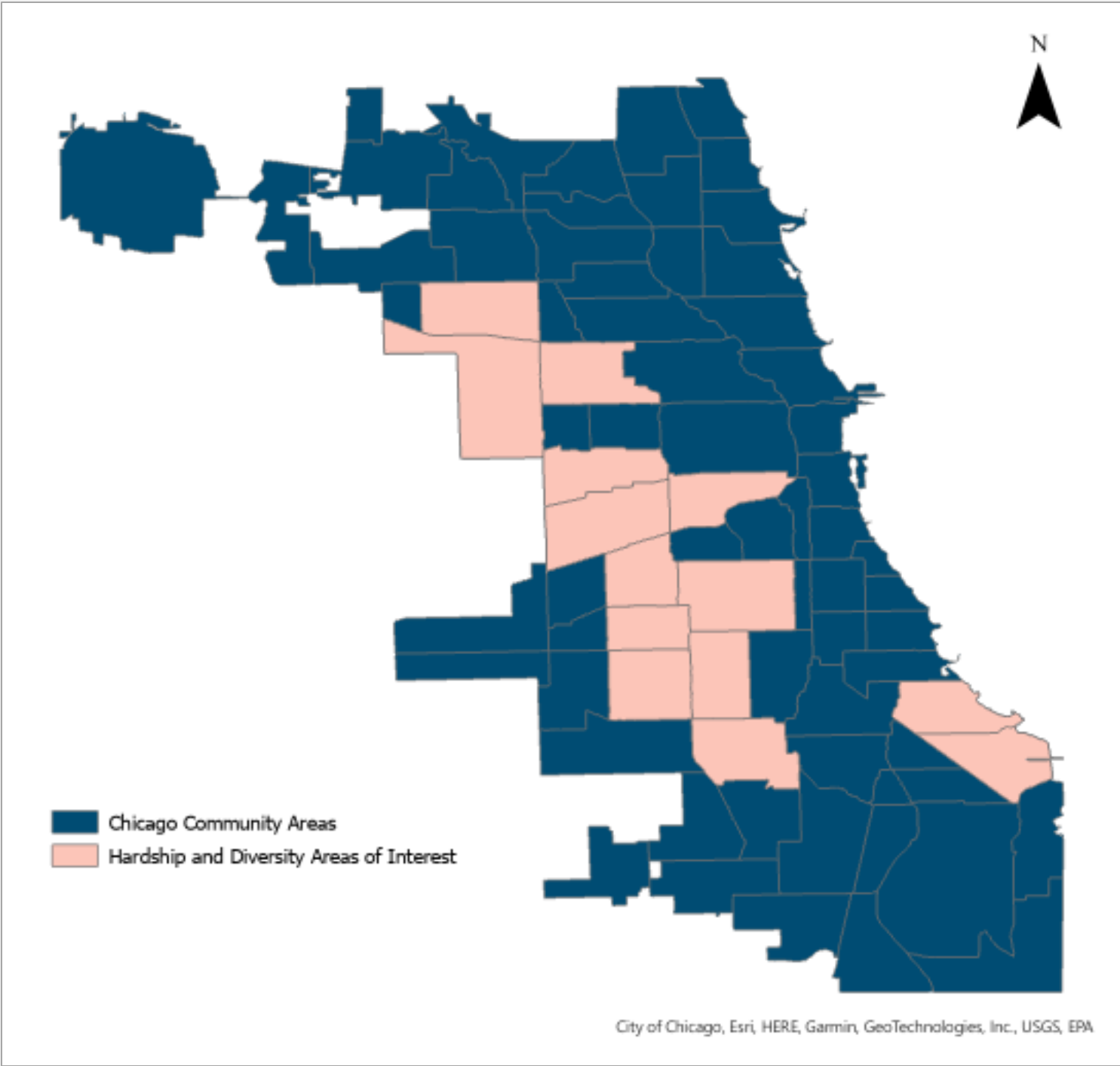
SQL

Pop_HardIx > %Hardship Index Threshold%

Used **Create Variable** to enable dynamic inputs for the **Index Escalator** and passed the resulting value into the **Calculate Value** tool. The output was then used to establish a **Hardship Index Threshold** for the **Select Layer by Attribute** tool.

Outputs

COMMUNITY1	PERC
Belmont Cragin	
Humboldt Park	
Austin	
North Lawndale	
South Lawndale	
Lower West Side	
South Shore	
South Chicago	
Brighton Park	
New City	
◀ ▶ 0 of 14 selected	



Section 4

Efficiency and Land Use – Submodel 3

Geoprocessing

←

Efficiency & Land Use Submodel

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Parameters

Environments

?

Suitable Zones

SuitableZones_%Land Use Class%

Chicago Efficiency

Chicago Efficiency

Land Use

Land Use

Efficiency Threshold Selector

Load Save Remove

←

→

✓

SQL

Where

gridcode

is ec

1

+

 Add Clause

▶

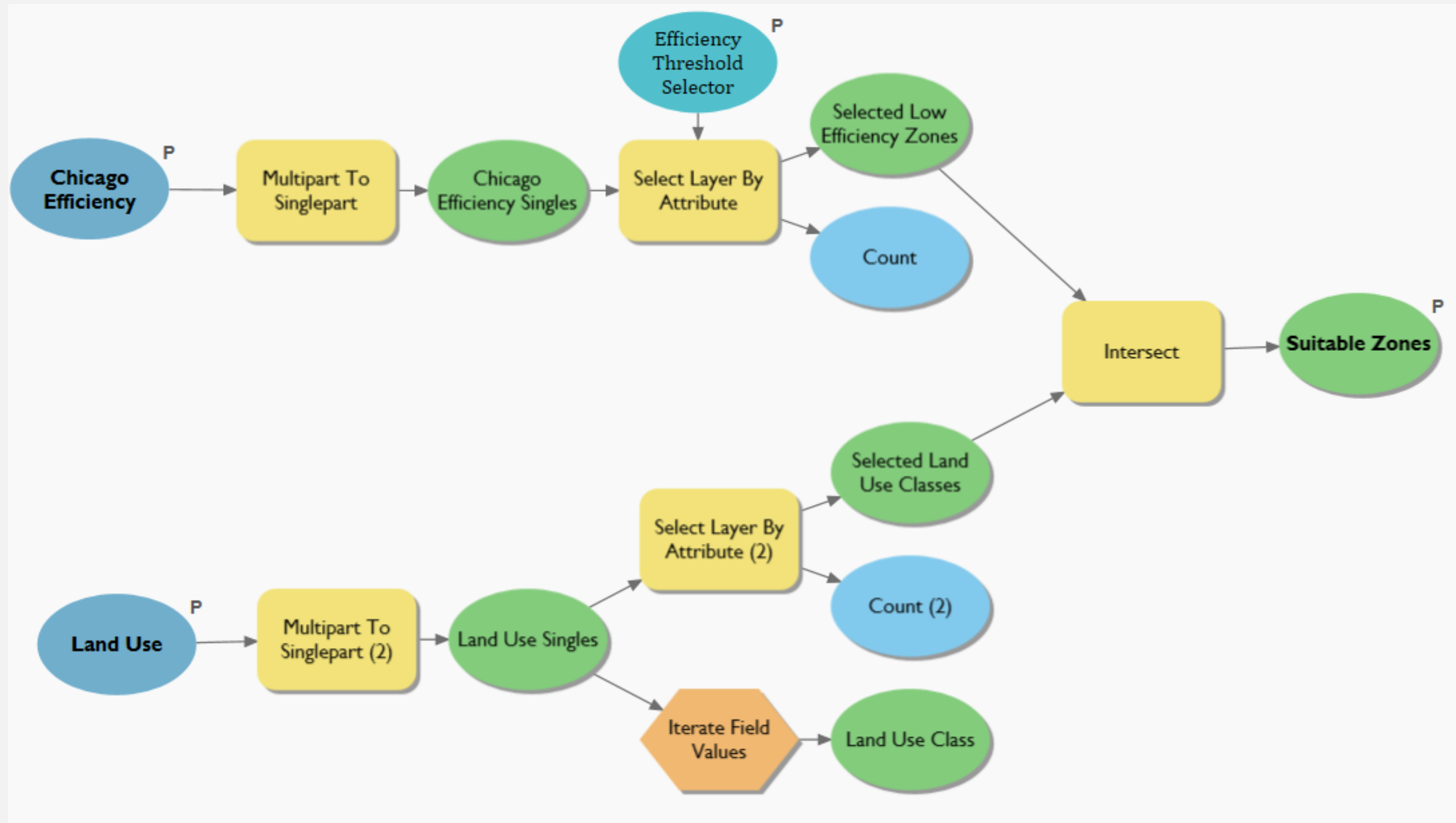
 Run

Overview

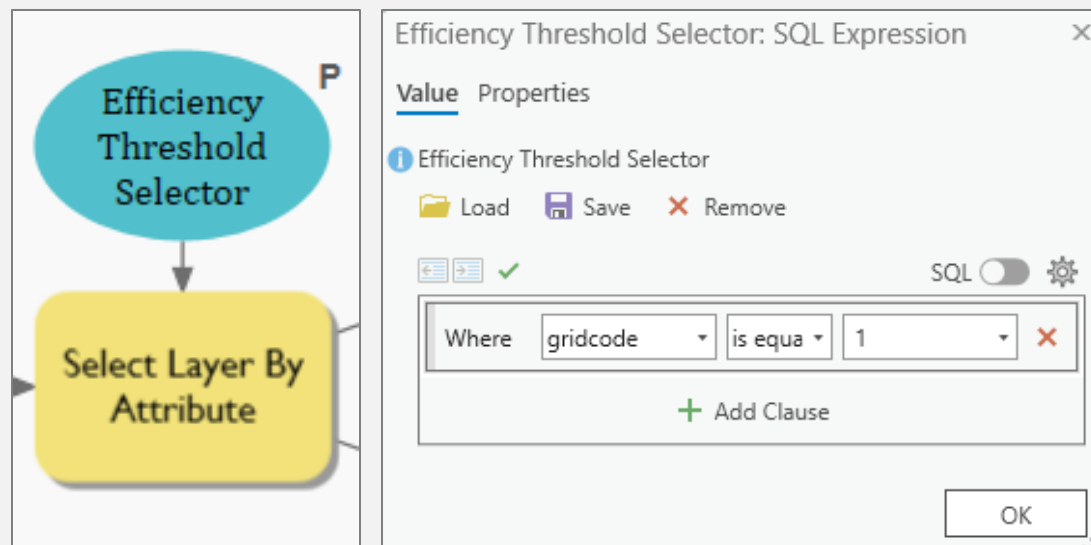
This submodel was not directly included in the supermodel but was designed to generate output feature layers that served as input layers in the supermodel. The submodel:

- Applied an **Iterator** to systematically process each land use type, ensuring all suitable zones were analyzed and outputs were dynamically generated.
- Used the Iterator to **Intersect** the low-efficiency zones with each land use type to identify areas that align with both energy and land use priorities.
- Generated **4 output layers** — suitable zones for Planned Development, Open Space/Residential, Business and Other land use classes.

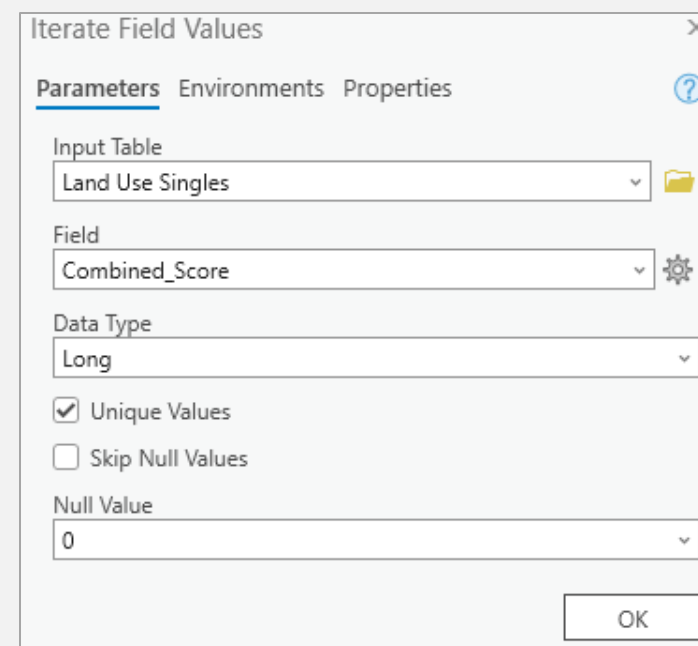
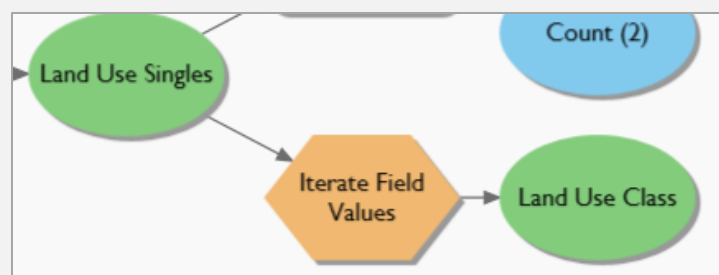
Overview



Key Features

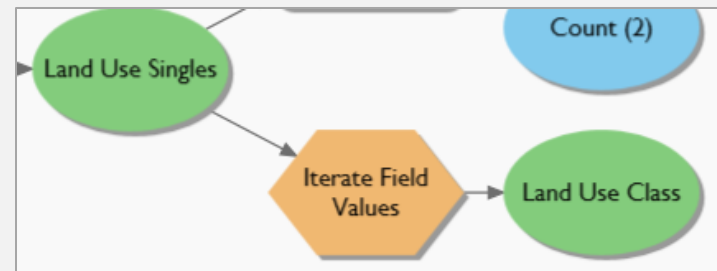
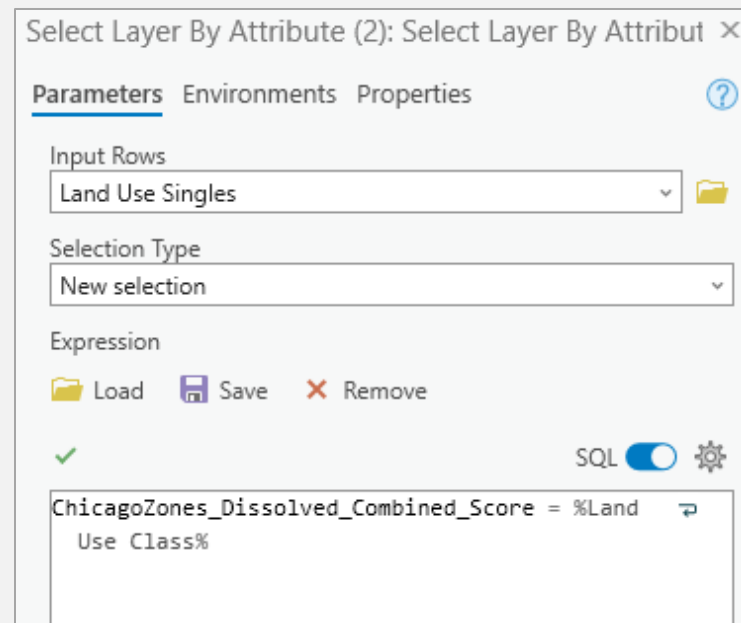


Created a **new variable** from the **expression parameter** of the **Select Layer by Attribute** tool to allow for modifications to include other levels of efficiency.



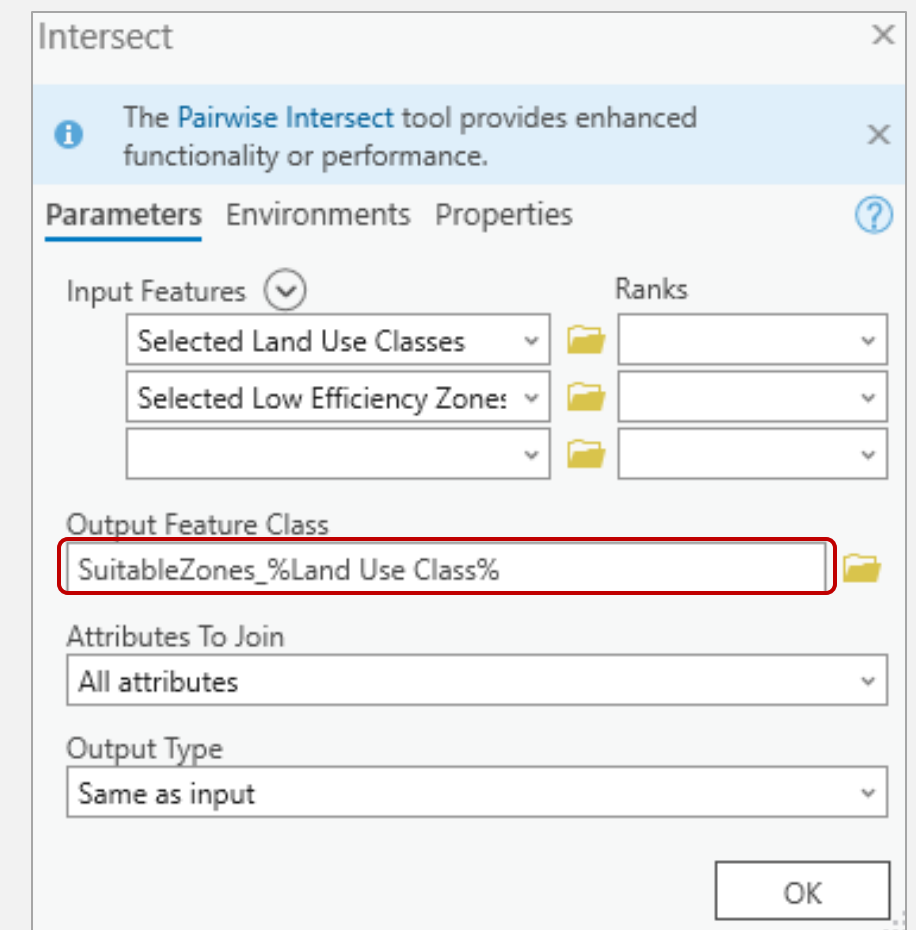
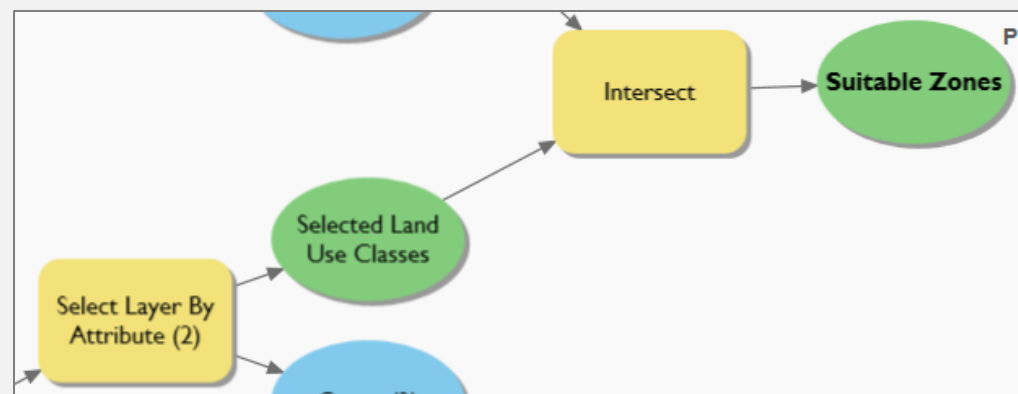
Used **Iterate Field Values** to enable each land use class to be processed separately in the subsequent selection and intersect operations. The iteration was based on the combined score with 1 representing "Others", 2 - "Business", 3 - "Open and Residential Spaces" and 4 - "Planned Development"

Key Features

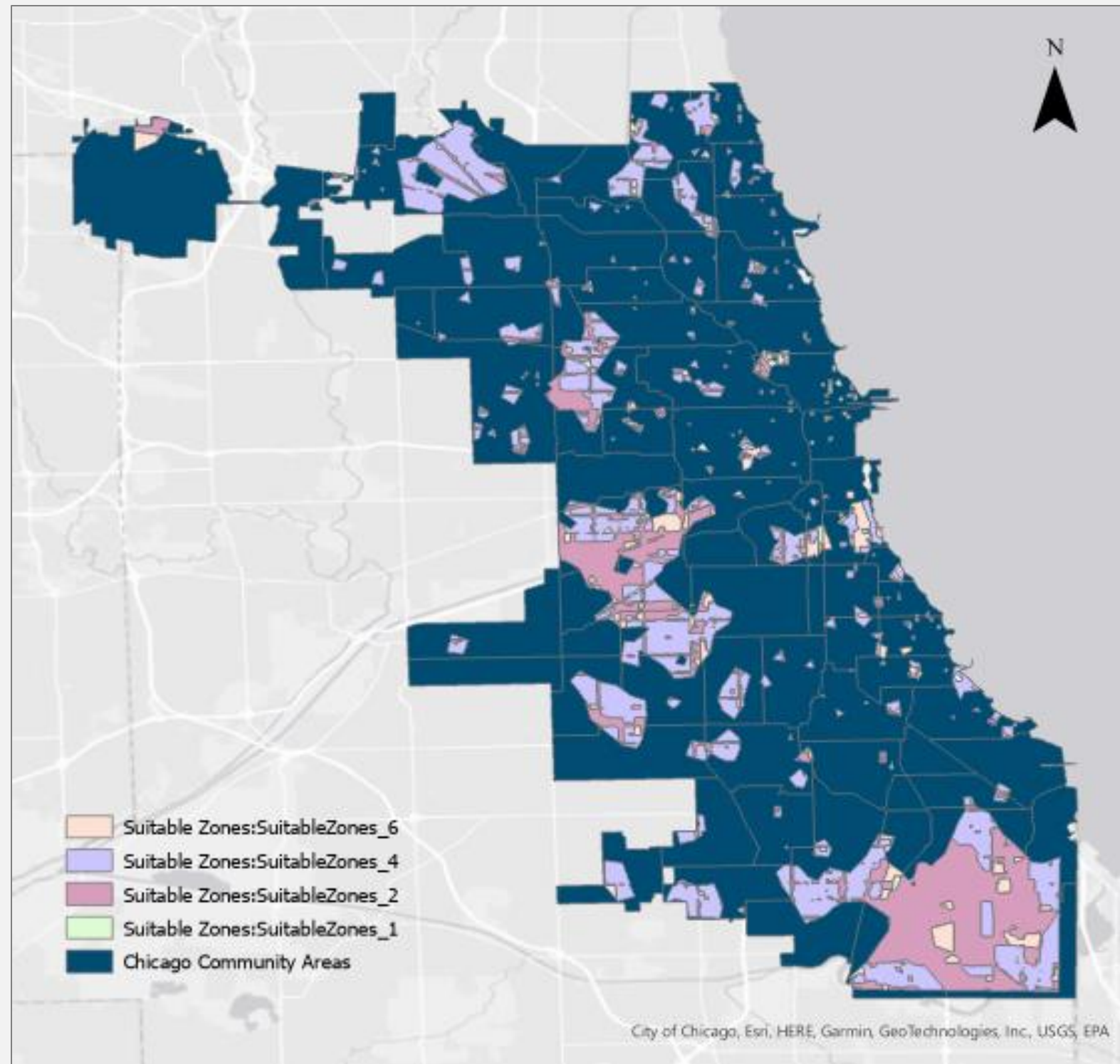


Used the output of the iterator – **Land Use Class** as the criterion for selection.

Calling for **%Land Use Class%** in the output feature class of the Intersect tool enabled the creation of **4 distinct feature layers** representing the intersection of each land use type and low efficiency zones



Outputs



Outputs

OBJECTID *	Shape *	FID_LandUseSingleparts	Combined_Score	Classification	ORIG_FID	FID_ChicagoEfficie_Singleparts
1	1	Polygon	3	1 Others	1	447
2	2	Polygon	4	1 Others	1	447
3	3	Polygon	5	1 Others	1	447
4	4	Polygon	6	1 Others	1	447
5	5	Polygon	7	1 Others	1	447
6	6	Polygon	8	1 Others	1	447
7	7	Polygon	9	1 Others	1	447
8	8	Polygon	10	1 Others	1	447
9	9	Polygon	11	1 Others	1	447
10	10	Polygon	12	1 Others	1	447

0 of 334 selected

OBJECTID *	Shape *	FID_LandUseSingleparts	Combined_Score	Classification	ORIG_FID	FID_ChicagoEfficie_Singleparts
1	Polygon	1322	2	Business	2	447
2	Polygon	1323	2	Business	2	447
3	Polygon	1324	2	Business	2	447
4	Polygon	1325	2	Business	2	447
5	Polygon	1326	2	Business	2	447
6	Polygon	1327	2	Business	2	447
7	Polygon	1328	2	Business	2	447
8	Polygon	1329	2	Business	2	447
9	Polygon	1331	2	Business	2	447
10	Polygon	1333	2	Business	2	447

0 of 433 selected

OBJECTID *	Shape *	FID_LandUseSingleparts	Combined_Score	Classification	ORIG_FID	FID_ChicagoEfficie_Singleparts
1	1	Polygon	2747	4 OpenSpace/ Residential	3	447
2	2	Polygon	2748	4 OpenSpace/ Residential	3	447
3	3	Polygon	2749	4 OpenSpace/ Residential	3	447
4	4	Polygon	2750	4 OpenSpace/ Residential	3	447
5	5	Polygon	2751	4 OpenSpace/ Residential	3	447
6	6	Polygon	2756	4 OpenSpace/ Residential	3	447
7	7	Polygon	2757	4 OpenSpace/ Residential	3	447
8	8	Polygon	2758	4 OpenSpace/ Residential	3	447
9	9	Polygon	2759	4 OpenSpace/ Residential	3	447
10	10	Polygon	2761	4 OpenSpace/ Residential	3	447

0 of 306 selected

OBJECTID *	Shape *	FID_LandUseSingleparts	Combined_Score	Classification	ORIG_FID	FID_ChicagoEfficie_Singleparts
1	1	Polygon	3235	6 Planned Development	4	447
2	2	Polygon	3236	6 Planned Development	4	447
3	3	Polygon	3237	6 Planned Development	4	447
4	4	Polygon	3238	6 Planned Development	4	447
5	5	Polygon	3239	6 Planned Development	4	447
6	6	Polygon	3240	6 Planned Development	4	447
7	7	Polygon	3241	6 Planned Development	4	447
8	8	Polygon	3242	6 Planned Development	4	447
9	9	Polygon	3244	6 Planned Development	4	447
10	10	Polygon	3247	6 Planned Development	4	447

0 of 258 selected

Section 5

Connecting the Dots

Geoprocessing

←

Supermodel

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Parameters

Environments

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Chicago Community Areas

Chicago Community Areas

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Diversity Analysis

DiversityAnalysis

▼

📁

Hardship Analysis

HardshipAnalysis

▼

📁

Suitable Zones: Open & Residential Areas

Suitable Zones:SuitableZones_4

▼

📁

Suitable Zones: Planned Development

Suitable Zones:SuitableZones_6

▼

📁

Sorted Total Score: Open/Res

Join_OpRes_Sort

▼

📁

Sorted Total Score: PlannedDev

Join_PlannedDev_Sort

▼

📁

▶ Run

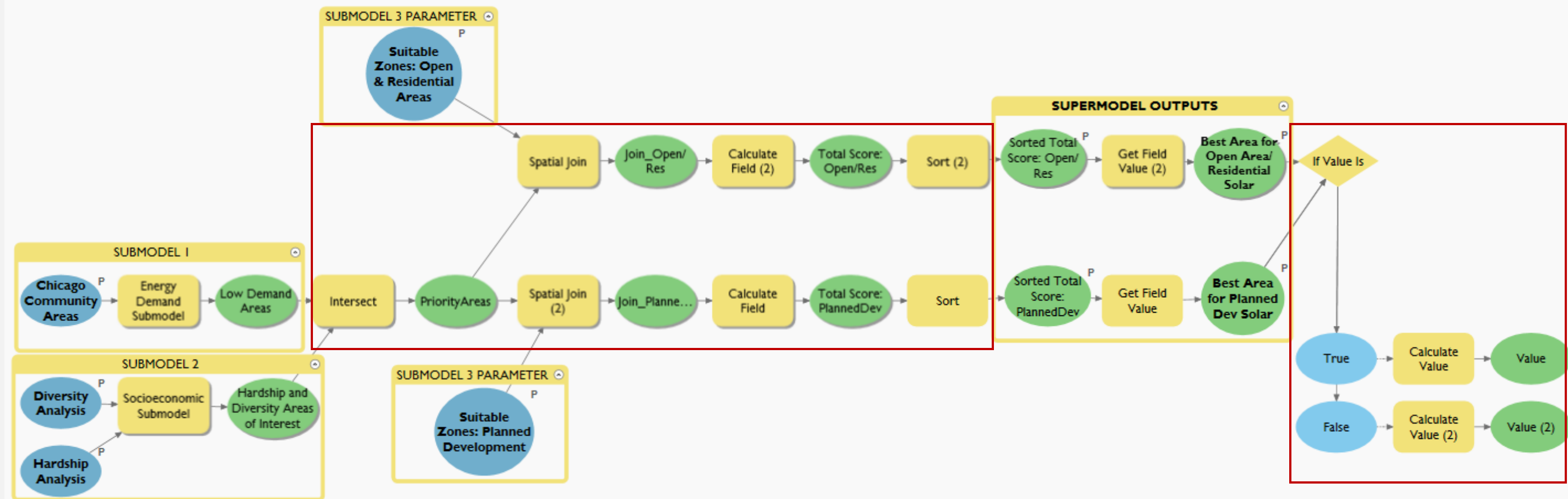
▼

Overview

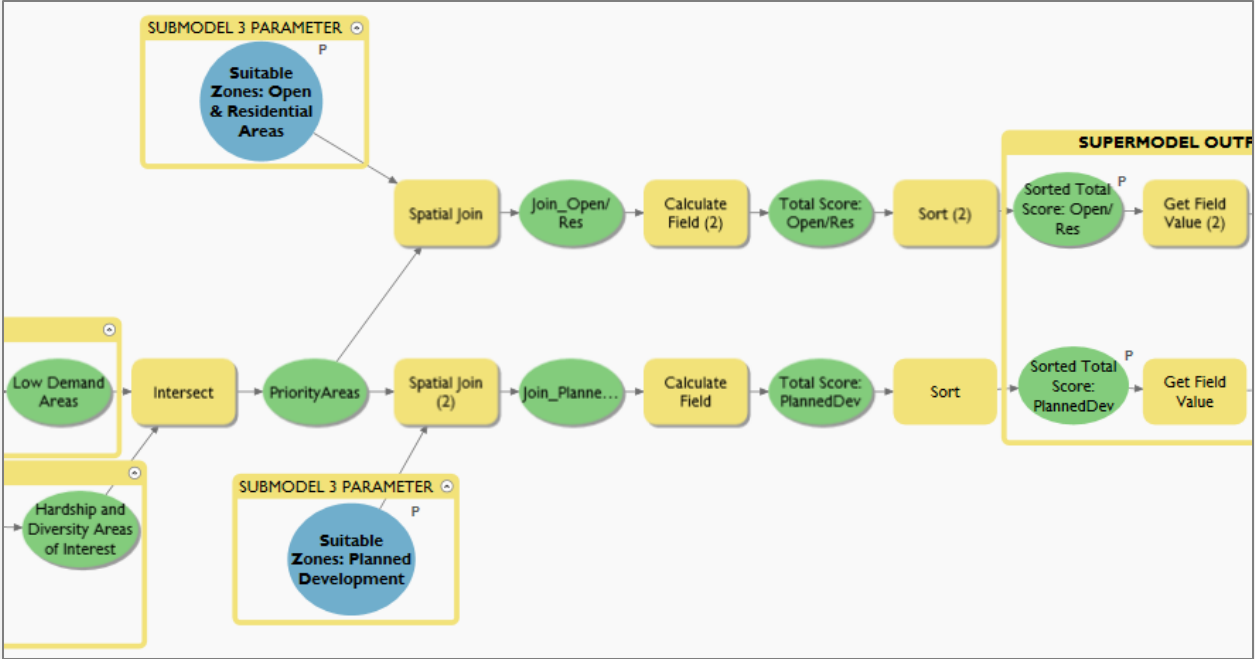
This supermodel integrates the three submodels to determine the most suitable towns for solar installations across two land use types: **Open/Residential Areas** and **Planned Development**. These land types were chosen because they have the highest suitability for solar installation. To connect the dots, the model:

- **Intersects** the outputs of Submodels 1 and 2
- **Spatially joins** this intersection with the Suitable Zones inputs derived from Submodel 3's outputs using the **Intersect match option**.
- **Calculates** a total score for each Community Area based on predefined criteria and sorts to rank the most suitable areas for each land type.
- Incorporates concluding **logic checks** to whether the two towns align.

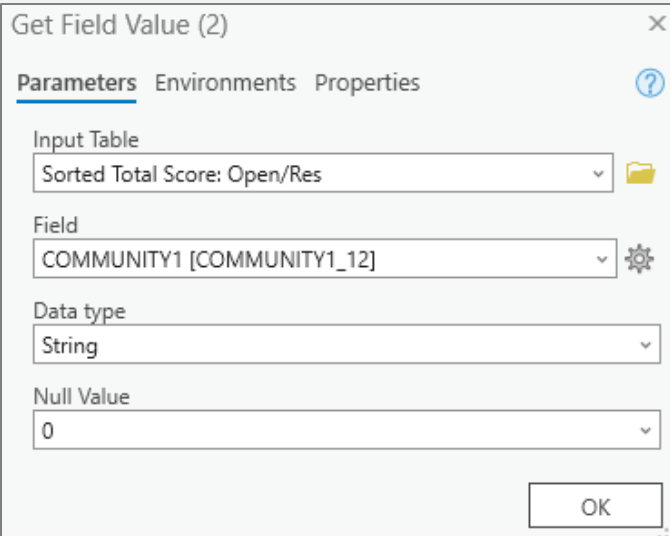
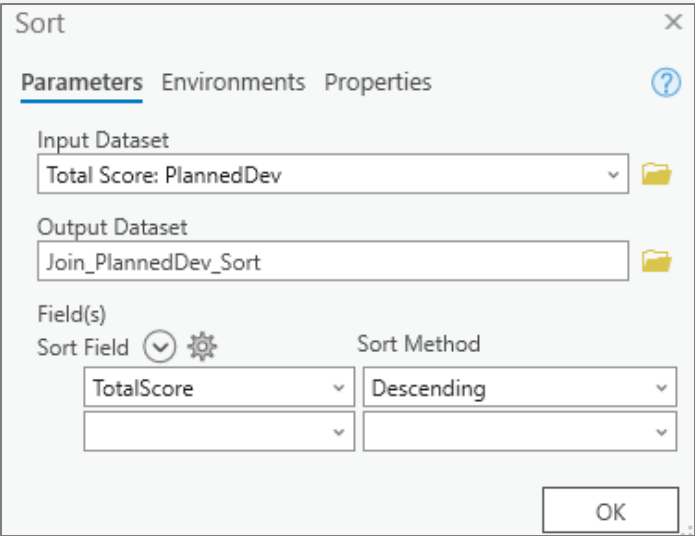
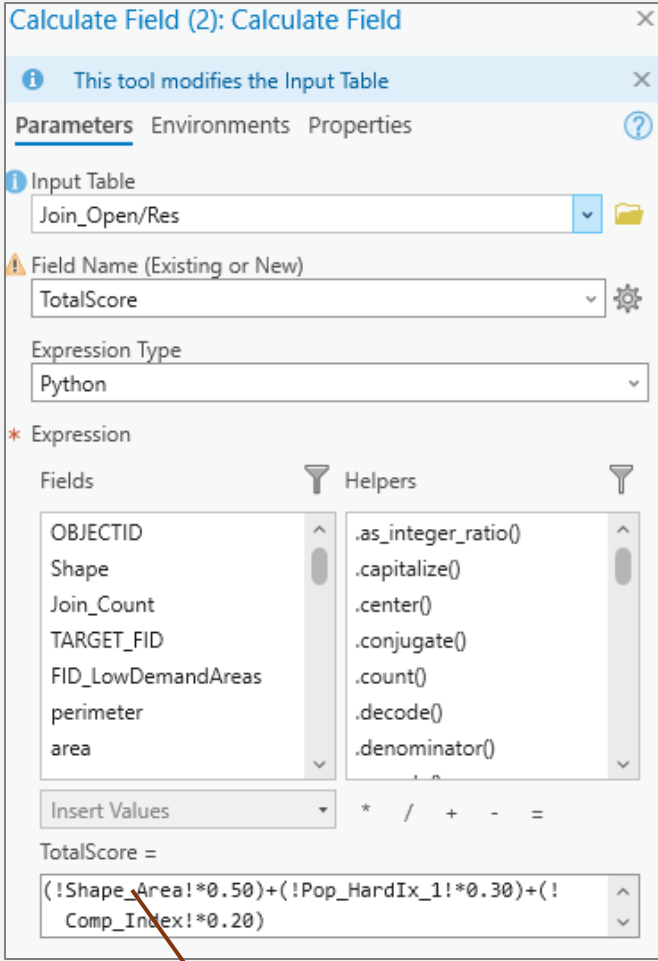
Overview



Key Features



Total Scores were computed with the **Calculated Field** tool using a formula that prioritizes the **Total Area of the Suitable Zones** in each **Community Area** for each land use class or type. The **Sort** tool arranged the total score in **descending order** to enable the **Get Field value** tool extract the **Community Area** in the first row.



TotalScore =
(!Shape_Area!*0.50)+(!Pop_HardIx_1!*0.30)+(!
Comp_Index!*0.20)

Key Features

If Value Is

Parameters

Environments

Properties

Input Value

Best Area for Open Area/Residential Solar

Value Test

Is equal to at least one value

Comparison Type

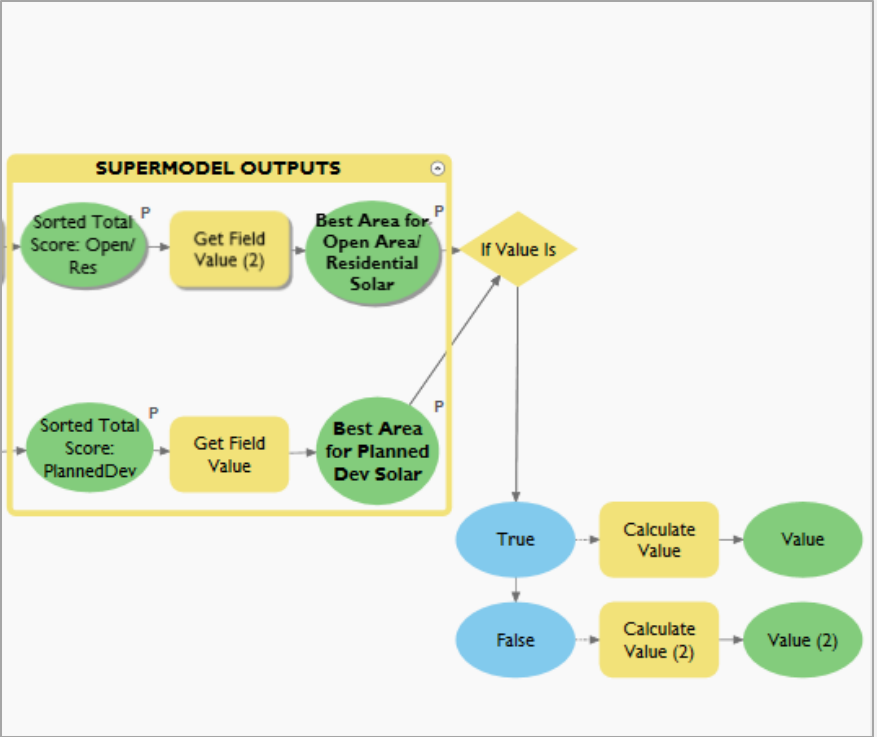
Case insensitive string

Values

Best Area for Planned Dev Solar

OK

The logical operator **If Value Is** compares the output from the sorted Community Area outputs to check whether both outputs are the same.



The logical operator feeds into the **Calculate Value** tool which extracts the name of the Best Community Area and returns the results of the logical check .

Calculate Value (2)

Parameters

Environments

Properties

Expression

Variables

Chicago Community Areas

Low Demand Areas

Diversity Analysis

Hardship Analysis

Hardship and Diversity Areas of Interest

PriorityAreas

Suitable Zones: Open & Residential Areas

Insert Values

Functions

.as_integer_ratio()

.capitalize()

.center()

.conjugate()

.count()

.decode()

.denominator()

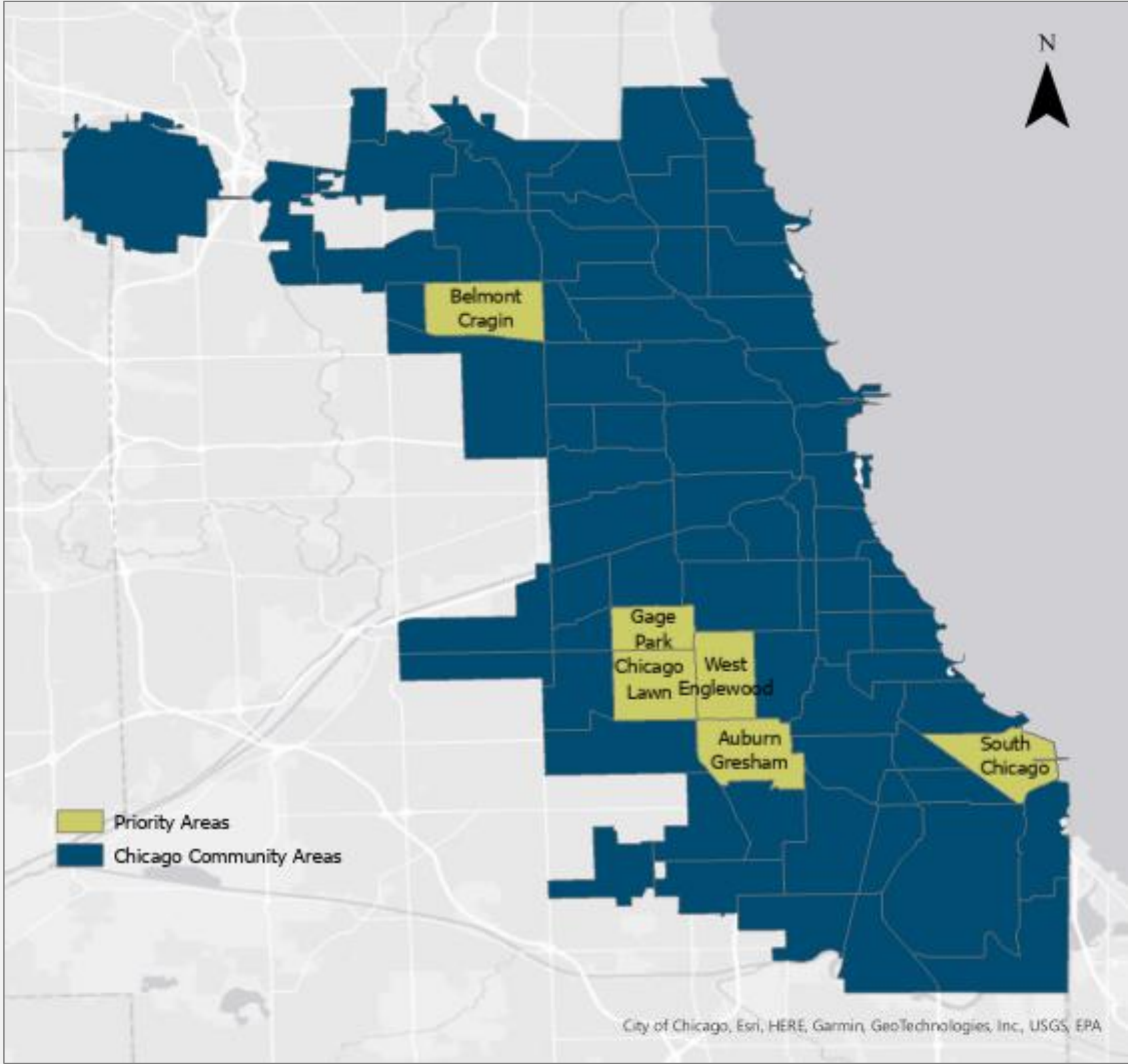
Expression

"%Best Area for Planned Dev Solar% and %Best Area for Open Area/Residential Solar% are ideal for Solar PV Installation in Planned Development and Open Space/

Expression

"%Best Area for Planned Dev Solar% is ideal for Solar Installation in both Planned Development and Open Space/

Outputs



Supermodel
Processing 14 of 14
Done (00:38,08)

Start Time: Sun
Time: 0.00 sec
The process
false.
Succeeded at Su
Time: 0.00 sec
Executing (Calc
Cragin and Gage
Planned Develop
respectively."
Variant
Start Time: Sun
Value = Belmont
Installation in
Space/Residenti
Succeeded at Su
Time: 0.00 sec

☐ Close on comp

Outputs

COMMUNITY AREA N...	TotalScore
Belmont Cragin	1553811.09239
Chicago Lawn	1259136.040765
Auburn Gresham	1031589.691741
Gage Park	970585.279832
West Englewood	688112.400442
South Chicago	682087.562088

Sorted Planned Development

COMMUNITY AREA N...	TotalScore
Gage Park	970585.279908369
West Englewood	688112.400442393
South Chicago	682087.562028913
Belmont Cragin	1553811.09239008
Chicago Lawn	1259136.04069434
Auburn Gresham	1031589.69168824

Sorted Open and Residential Areas

Outputs

Supermodel

Processing 14 of 14

Done (00:38.08)

Start Time: Sunday, December 15, 2024 8:25:46 AM

⚠ The process did not execute because the precondition is false.

Succeeded at Sunday, December 15, 2024 8:25:46 AM (Elapsed Time: 0.00 seconds)

Executing (Calculate Value (2)): CalculateValue "Belmont Cragin and Gage Park are ideal for Solar PV Installation in Planned Development and Open Space/Residential Zones respectively."

Variant

Start Time: Sunday, December 15, 2024 8:25:46 AM

Value = Belmont Cragin and Gage Park are ideal for Solar PV Installation in Planned Development and Open Space/Residential Zones respectively.

Succeeded at Sunday, December 15, 2024 8:25:46 AM (Elapsed Time: 0.00 seconds)

☐ Close on completion

Value (2): Any Value

Value (2)

Belmont Cragin and Gage Park are ideal for Solar PV Install.

OK

Section 4

Conclusion and References

Conclusion

By designing key variables as parameters, the model allows users to adjust inputs and thresholds, enabling experimentation and adaptability for different scenarios.

The analysis identified **Gage Park** as the best town for **Open/Residential Areas** and **Belmont Cragin** as the best for **Planned Development**. Notably, Belmont Cragin aligns with the single community area highlighted in the Data Interpretation Project.

References

- <https://greatcities.uic.edu/wp-content/uploads/2019/12/Hardship-Index-Fact-Sheet-2017-ACS-Final.pdf>
- <https://data.cityofchicago.org>
- <https://cmap.illinois.gov>
- https://www.aceee.org/sites/default/files/pdfs/aceee-01_energy_burden_-_chicago.pdf?utm_source=chatgpt.com
- https://chicagopolicyreview.org/2022/01/12/advancing-energy-equity-and-climate-solutions-in-chicago/?utm_source=chatgpt.com